Controller–Host Agreement on Live Research Decisions: A Receipted Benchmark and First Measurement

Anonymous authors

August 2026

Abstract

Research-agent benchmarks usually score decisions against known answers or environment rewards. Live research questions are different: the correct next move may not yet have a gold label. We define a controller–host agreement benchmark for this setting. Two architecturally distinct instruments receive the same frozen evidence packet and frontier question. One is an LLM-host-driven research lane whose external interactions are bound to immutable capability receipts; the other is a typed campaign controller whose decision path contains no LLM and whose revision/computation choices are constrained by explicit obligations. The benchmark scores whether the lanes diagnose the same responsible layer, select compatible next moves, and use overlapping admissible evidence, with a later research outcome available as deferred rather than contemporaneous scoring. Benchmark V0 was frozen before either lane outcome. Both instruments diagnosed the representation-regime-characterization layer and selected regime characterization as the primary move. The typed controller additionally withheld an unlicensed representation split because its membership-predicate obligation remained unresolved. Later characterization and closure results make the deferred coordinate 'ALIGNED'. This is a benchmark definition with one measurement, not an evaluation of reliability or predictive validity. The broader systems-harness guarantee surface is outside this paper and is routed to separate work. This report presents a first measurement on a single frontier question; additional instances are required before the agreement measure can support a general claim.

Keywords: research agents; benchmark; agreement; epistemic control; provenance; scientific workflows.

1 Introduction

When a research system is asked a question whose answer is already known, task success is the natural endpoint. A frontier decision poses a different measurement problem. The system may need to decide which assumption failed, which representation should change, or which experiment should run next before ground truth exists. Repeated sampling from one model can measure self-consistency, but it does not provide an architecturally distinct instrument that can disagree for different reasons.

This benchmark defines a benchmark around that gap. The object measured is not whether an LLM is correct and not whether a typed controller is correct. It is whether two different decision procedures, constrained to the same admitted evidence, converge on the same epistemic diagnosis and next action under a protocol frozen before either outcome. Divergence is data, not failure.

The first lane is host-driven and can use an LLM to synthesize a decision, but its evidence-bearing external interactions are captured in content-bound receipts. The second lane uses typed production control modules over a frozen manifest and receipt-transcribed observations; no LLM participates in its decision path. The distinction is architectural rather than absolute: both lanes are constructed inside one programme and share the evidence corpus by design.

Benchmark V0 asks what layer is responsible after a sequence of exact TARE-family closure/refutation results and what move should come next. The result is agreement. The paper reports that measurement and its decision mechanics while resisting the tempting but unsupported next sentence: one aligned instance does not establish instrument reliability.

2 Methods

2.1 Benchmark construct

The V0 protocol freezes a single live frontier question, the admissible receipt corpus, the two lane definitions, scored coordinates, tie/compatibility rules, and the interpretation of divergence before either lane result exists. Scoring separates (1) responsible-layer diagnosis, (2) selected next move, (3) evidence overlap/admissibility, and (4) deferred alignment with later research outcomes.

2.2 Lane A: host-driven instrument

Lane A operates through a generic research harness. Capability requests have deterministic identities derived from canonicalized request content, and results are digest-bound to the exact request. The host may use an LLM for synthesis, but the benchmark answer must be reconstructable from admitted receipt-backed evidence. The lane's free-text reasoning is therefore not itself an authority source.

2.3 Lane B: typed controller

Lane B receives a frozen campaign manifest containing competing responsibility hypotheses, expected observations, revision mechanics, obligation states, and computation actions. Production epistemic-control modules select a hypothesis and action over typed observations. The decision path contains no LLM. State, decision, and transition records serialize authority grants as false.

2.4 Revision gate

The surviving representation-regime hypothesis is associated with a representation-split revision that requires a frozen regime-membership predicate. At the V0 decision point that obligation is unresolved. Under fail-closed controller semantics the revision is unavailable, leaving characterization computation as the admissible next move.

2.5 Independence and deferred scoring

The lanes are independent in decision procedure but not in evidence source: both use the same committed R6 receipt corpus. Lane B's manifest was authored/transcribed by a human/LLM process before the native decision, so framing sensitivity remains a possible dependence channel. Deferred scoring uses later research outcomes as a diagnostic alignment signal, not as an externally supplied gold label.

3 Results

3.1 V0 agreement

Lane A identifies the responsible layer as representation-regime characterization and selects regime-predicate characterization as primary, with the support-two closure test as a complementary probe. Its answer is bound to 13 host capability receipts and the verified evidence items admitted by the frozen protocol.

Lane B identifies the corresponding typed responsibility state, defeats the current-search, donor-family, and method-language alternatives, and selects 'COMPUTE:REGIME_CHARACTERIZATION'. Its native decision is bound to the frozen manifest and transition receipts. The revision gate reports the representation split as unresolved because the required predicate obligation has not yet been satisfied.

Under the protocol's compatibility rules the contemporaneous verdict is 'AGREE': the diagnosis and primary move match at the scored level, and the evidence sets overlap by construction while remaining receipt-bound.

3.2 Deferred coordinate

After V0, the independently launched frontier lanes produce an exact regime-membership predicate on their stated domains and restore family closure at support two, the two directions prioritized by Lane A and structurally licensed by Lane B. Coordinate 4 is therefore recorded 'ALIGNED'. This retrospective score is descriptive; it does not turn the subsequent result into a gold label for all frontier decisions.

3.3 Historical defects are repaired

The V1 manuscript documented two real harness defects: malformed successful reasoner content could escape as an unstructured crash, and the deterministic request identity could then be pinned by an immutable semantically invalid receipt. Current regression tests now require structured 'HOST_CAPABILITY_FAILED' behavior and provide an explicit, reason-bearing invalid-content archival path that preserves the original bytes before allowing a corrected receipt. These repairs update the implementation status but do not add benchmark instances.

3.4 Evidence-series gate

No V1/V2/V3 benchmark series is present on current main. The result set therefore remains exactly one scored question. We report no agreement percentage, confidence interval, calibration measure, or predictive-validity statistic.

4 Discussion

V0 demonstrates that controller–host agreement can be made into a replayable measurement rather than an anecdotal statement that two systems 'thought the same thing'. The receipt layer fixes what evidence each lane used; the typed controller exposes why one revision was unavailable; and the frozen scoring contract makes divergence reportable.

The most informative aspect of the agreement is not identical wording. Lane A reaches characterization through evidence synthesis. Lane B reaches it because the alternative revision lacks a resolved obligation. Agreement across these decision mechanics is potentially more informative than repeated samples from one substrate, but V0 alone cannot quantify how much more informative.

The benchmark should also be expected to produce disagreement. Useful future instances should include cases where one lane detects an evidence/authority obstruction the other misses, where both select wrong-but-different moves under later scoring, and where manifest framing changes the typed controller's decision. A benchmark series that only selects questions likely to agree would invalidate the construct.

The systems contribution is intentionally out of scope. separate work is the place to evaluate receipt publication, bounded outputs, process safety, recovery semantics, and the relationship between the two harnesses as execution systems. This benchmark uses those mechanisms only to make the agreement measurement auditable.

5 Related work and boundary

AgentBench evaluates language-model agents across interactive environments with task-level outcomes [1], and SWE-bench evaluates language models against real software issues with repository-grounded success criteria [2]. Autonomous-science systems such as Coscientist demonstrate tool-using LLM workflows for planning and experimental execution [3]. Provenance research supplies a separate conceptual parent for tracing derived decisions to source records [4]. These works support the neighboring benchmark, agent, and provenance contexts; none is cited as evidence that agreement between two instruments is itself predictive of scientific correctness.

Agreement as evidence, and when it is not. The strongest challenge to any agreement-based measurement comes from work on judge consensus. Mukherjee et al. show that LLM judges agree substantially more with one another than with humans, and that this consensus is geometric rather than epistemic: judges collapse onto a shared low-rank subspace whose principal angle to the human subspace stays near orthogonal even after fine-tuning and preference optimisation [5]. Their conclusion is that inter-judge agreement should count as evidence of alignment only when a direct structural check on the judges' shared subspace passes. Their own diagnosis identifies the mechanism: within-family covariance is the strongest orientation signal, so shared pre-training and post-training data, not scale, drive the collapse.

That mechanism is the reason this benchmark pairs architecturally distinct instruments rather than two model instances. One lane is LLM-host-driven; the other contains no language model anywhere in its decision path, and its choices are constrained by explicit obligations rather than learned preferences. Two lanes that share no pre-training corpus cannot agree by collapsing onto a corpus-frequent direction, because only one of them has a corpus. This does not make agreement sufficient evidence of correctness, and we do not claim it does; it removes the specific confound that makes homogeneous-judge consensus uninformative.

The no-gold-label setting carries a second known hazard. Kranti and Vajjala show that judges over-credit incorrect answers when no reference is available, and that supplying a reference flips a large fraction of verdicts [6]. Their result argues against treating any single reference-free verdict as a correctness signal. The design here responds by never scoring a lane's answer as correct in the absence of ground truth: agreement is recorded as agreement, divergence is an admissible outcome rather than a failure, and correctness is assigned only later, by deferred scoring once a research outcome exists.

Boundary. The bounded candidate residual is the specific measurement contract: the same frozen frontier question, architecturally distinct decision paths, receipt-bound evidence, divergence as an admissible outcome, and deferred scoring when contemporaneous ground truth is unavailable. That residual remains conditional on a fresh submission-date search; this paper therefore does not claim that no prior benchmark measures any form of cross-instrument agreement.

The broader host and capability-receipt architecture is not claimed as a standalone systems contribution of this paper. That subject requires its own donor matrix and protocol before it can carry a systems-paper claim, and is deferred to separate work.

Why architectural distinctness is the design variable. The claim that agreement between similar systems carries little information has been tested directly. Ji evaluates a three-agent panel on six groundedness benchmarks and finds the system-level difference against a single-agent reference ranges from +8.5 to −4.4 points, positive on two datasets, negative on one, and inconclusive on three [7]. The mechanism analysis is the part that matters here. Second-round arguments carry low semantic novelty; the accuracy changes decompose into class-specific recall shifts rather than new evidence; and persona errors are strongly correlated, reaching $\phi = 0.93$ between two of the three roles. The paper's own summary of the limit is that agents sharing a model, evidence pool and fixed aggregation rule can move a decision boundary without acquiring independent evidence, and it closes by naming model or evidence diversity as a precondition for reliable gains.

That is the design variable this benchmark manipulates, and it is manipulated further than the diversity Ji recommends. Two lanes here do not merely differ in prompt or persona; one contains no language model at any point in its decision path, and its choices are constrained by explicit obligations rather than learned preferences. There is no shared pre-training corpus to collapse onto, because only one lane has a corpus at all. Ji also supplies a caution this benchmark adopts rather than argues for: a changed answer is not a corrected answer — on one dataset only 29.6% of label flips landed on the correct label. Agreement is therefore recorded here as agreement, divergence is an admissible outcome, and correctness is assigned separately and later.

6 Limitations

The decisive limitation is sample size: V0 is one frontier question. No reliability, agreement-rate, predictive-validity, or calibration claim follows. This is also a publication blocker under the prospectively recorded this benchmark plan, which requires at least two to three additional instances.

The lanes are not fully independent. They share the receipt corpus, originate in the same research programme, and Lane B's manifest is human/LLM-authored even though the native decision is not. Manifest construction can encode framing choices. Lane A's LLM synthesis may also share conceptual priors with the people who designed the typed hypotheses.

Deferred scoring is endogenous to the same programme and is not external ground truth. The current benchmark does not establish that agreement predicts scientific success, only that the chosen V0 direction was later aligned with the programme's own frozen frontier outcomes.

Finally, reproducible receipts do not make a decision scientifically correct. They make the evidence/decision path attributable. That work's systems guarantees and this paper's methodological claims are therefore distinct from this paper's benchmark interpretation.

7 Conclusion

Controller–host agreement provides a measurable object for live research decisions that do not yet have a gold label. In the frozen V0 instance, an LLM-host-driven lane and a typed non-LLM controller independently identify the same responsible layer and compatible next move, while the typed revision gate exposes why a stronger representation change is not yet licensed. Later frontier outcomes make the deferred coordinate aligned.

The result should remain exactly that small. V0 demonstrates the benchmark contract and one reproducible measurement. A standalone evaluation paper requires the additional prospectively frozen instances already specified by the programme; until they exist, the benchmark's reliability remains an open empirical question.

8 Reproducibility

The V0 protocol, lane receipts, result JSON, and workspace archive are committed under `development/orion-q-max-r0/`. Lane A can be audited by recomputing request/result identities and tracing the final synthesis to its verified evidence items. Lane B can be replayed from the frozen manifest and receipt-transcribed observations; a manifest-digest mismatch must fail before a scientific decision is accepted.

Deferred scoring is independently reconstructable from the committed R6P/R6Q result artifacts. Historical invalid-content recovery should be tested with `packages/orion-research-harness/tests/test_invalid_content_recovery.py`.

The root `REPRODUCE.md` gives the intended verification order.

The missing benchmark series is not a reproducibility failure. It is missing evidence and requires new pre-outcome protocols/executions.

9 Ethics, safety and resource statement

No human participants, animals, clinical data, or personal data are used. The main integrity risks are authority laundering, hidden evidence asymmetry between lanes, post-outcome scoring changes, and selective reporting of agreement. The protocol addresses these by freezing evidence/scoring before outcomes, making divergence an admissible terminal, and recording all authority grants as false in the typed lane.

The benchmark should not be used to claim that consensus is correctness. Future series must retain disagreements and adverse deferred outcomes. Receipt coverage is provenance, not evidence quality, and neither lane is permitted to turn a host/tool failure into scientific evidence.

References

[1] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. AgentBench: Evaluating LLMs as agents. In *The Twelfth International Conference on Learning Representations*, 2024.

[2] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *The Twelfth International Conference on Learning Representations*, 2024.

[3] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023.

[4] Peter Buneman, Sanjeev Khanna, and Wang-Chiew Tan. Why and where: A characterization of data provenance. In *Database Theory—ICDT 2001*, volume 1973 of *Lecture Notes in Computer Science*, pages 316–330. Springer, 2001.

[5] Sourabrata Mukherjee, Hamna, Kalika Bali, and Sunayana Sitaram. The geometry of LLM-as-judge: Why inter-LLM consensus is not human alignment. 2026.

[6] Chalamalasetti Kranti and Sowmya Vajjala. LLM judges can be too generous when there is no reference answer. 2026.

[7] Yuelyu Ji. More debate, same evidence: Structural limits of homogeneous multi-agent groundedness judges. 2026.